

CSA402 - CO4 - ATJ compiler design

(1) construct the address code

if $((a < b) \& (c < d) \text{ or } (a > d))$ then

$z = x + y * z$

else

$z = z + 1$

Given

if $((a < b) \text{ and } ((c < d) \text{ or } (a > d)))$ then

$z = x + y * z$

else

$z = z + 1$

expressions: $a < b$

$c < d$

$a > d$

loops: L1, L2, L3, L4, L5, L6

L1:

if $a < b$ goto L2
goto L5

L2:

if $c > d$ goto L4
goto L3

L3:

if $(a > b)$ goto L4
goto L5

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L4: $z = x + y * 2$

L4 $x1 = y * 2$

$x2 = x + x1$

$z = x2$

goto L6

L5: $z = z + 1$

L5:

$x3 = z + 1$

$z = x3$

2) Translate $(a < b) \& (c < d) \text{ or } (c < f)$ into three address statements using Backpatching

$[(a < b) \text{ and } (c > d)] \text{ or } (c < f)$

if $(a < b)$ goto 3

goto 5

if $(c < d)$ goto 11

goto 5

if $(c < f)$ goto 7

goto 8

true

false

Construct three address code use the following production for boolean $(a > b) \text{ or } (c < d) \text{ or } (c < f)$

$B \rightarrow B_1 \parallel B_2$

$B \rightarrow B_1 \&\& B_2$

$B \rightarrow !B_1$

$B \rightarrow E_1 \text{ relop } E_2$

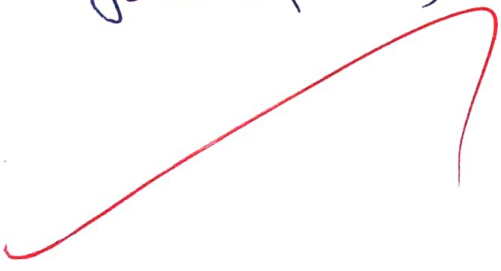
$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

L1: if $(a > b)$ goto Ltrue
goto L2

L2: if $(c < d)$ goto LTRUE
goto L3

L3: if $(c < f)$ goto LTRUE
goto [false]



(4) generate the three address code for following program fragment where $(A < C \ \& \ B < D)$ do

if $A = 1$ then $C = C + 1$
else while $A <= 0$ do
 $A = A + B$

L1:
 if $A < C$ goto L2
 goto LEND

L2:
 if $B > D$ goto L3
 goto LEND

L3: if $A == 1$ goto L4
 goto L5

L4:
 $t1 = C + 1$
 $C = t1$
 goto L1

L5: if $A <= 0$ goto L6
 goto L1

L6: $t2 = A + B$
 $A = t2$
 goto L5

LEND: